

What is UNIX?

An operating system claimed to be “standard” “from microcomputers to supercomputers”

Invented by AT&T Bell Labs in late 60's

Currently there are different versions of UNIX such as

- SunOS,
- Linux,
- DEC OSF/1,
- AIX,
- HP-UX 10,
- Solaris,
- IRIX,
- SCO UNIX, etc.

Important directories in UNIX

- /home - all users. home directories are stored here
- /bin, /usr/bin - system commands
- /sbin, /usr/sbin - commands used by sysadmins
- /etc - all sorts of configuration files
- /var - logs, spool directories etc.
 - /var/log The log-files of the system.
 - /var/spool Directories for mail, news, printing and other queued work.
- /dev - device files
- /proc - special system files
- /boot - kernel image and configuration files used by boot loader
- /tmp - location of temporary files
- /mnt - mount point for temporarily mounting files system such as those on a CDROM or floppy disk
- /opt - add-on application software packages
- /root - the root user's home directory
- /usr - Everything related to users
- /usr/bin - houses critical binaries of the users
- /usr/include - the header-files required by programs for compilation.
- /usr/lib - The libraries required by user-applications.
- /usr/sbin - User-administration binaries.

These are at Linux Servers

- /usr/src The source-code for the Linux kernel.
- /usr/X11R6 Files needed by the X Window system.

SOME UNIX COMMAND			
ls	• sort	▪ su	■ sed
• cp	• head	▪ file	■ tar
• mv	• more	▪ diff	■ awk
• chown	• cat	▪ passwd	■ cpio
• chgrp	• tail	▪ su	■ kill
• clear	• grep	▪ wc	■ type
• who	• cd	▪ du	■ which
• w	• pwd	▪ df	■ less
• uname	• rm	▪ uniq	■ cut
• man	• mkdir	▪ echo	■ jobs
• whereis	• touch	▪ alias	
• find			

Login to the System

login as: mpn00221

Keyboard-interactive authentication prompts from server:

| Password:

End of keyboard-interactive prompts from server

Last login: Wed Jul 6 14:03:07 2022 from 10.90.20.33

p01vspplmapp01 mpn00221 [~] \$

p01vspplmapp01 mpn00221 [~] \$

p01vspplmapp01 mpn00221 [~] \$ **who am I (Detail of user)**

mpn00221 pts/3 2022-07-07 09:26 (10.90.20.33)

p01vspplmapp01 mpn00221 [~] \$ **id** (User Information – user Identification and group details)

uid=**1002**(**mpn00221**) gid=**1000**(**dba**) groups=1000(dba)

p01vspplmapp01 mpn00221 [~] \$ **who -a** (who are currently available in the system)

system boot 2022-06-03 09:51

(unknown) ? :0 2022-06-03 09:51 ? 2006 (:0)

run-level 5 2022-06-03 09:51

LOGIN tty1 2022-06-03 09:51 1674 id=tty1

tcuaprd + pts/0 2022-06-14 18:15 old 97882 (10.90.20.38)

tcuaprd + pts/1 2022-06-03 09:55 old 3344 (10.90.20.36)

tcuaprd + pts/2 2022-06-08 15:23 19:43 61500 (10.90.20.38)

mpn00221 + pts/3 2022-07-07 09:26 . 61478 (10.90.20.33)

tcuaprd + pts/4 2022-06-13 12:31 old 86969 (10.90.20.38)

pts/5 2022-07-06 17:11 102713 id=/5 term=0 exit=0

tcuaprd + pts/6 2022-06-15 14:26 old 62662 (10.90.20.36)

pts/7 2022-06-30 11:29 64708 id=/7 term=0 exit=0

tcuaprd + pts/8 2022-06-14 19:11 old 102092 (10.90.20.38)

tcuaprd + pts/9 2022-06-14 19:13 old 102328 (10.90.20.38)

tcuaprd + pts/10 2022-06-14 19:14 old 102516 (10.90.20.38)

tcuaprd + pts/11 2022-06-15 11:19 old 47015 (10.90.20.36)

pts/12 2022-06-30 11:29 72601 id=/12 term=0 exit=0

pts/13 2022-06-30 11:29 74733 id=/13 term=0 exit=0

pts/14 2022-06-30 11:29 76934 id=/14 term=0 exit=0

pts/15 2022-06-23 09:46 31833 id=/15 term=0 exit=0

pts/16 2022-06-22 16:04 10511 id=/16 term=0 exit=0

pts/17 2022-06-22 16:04 10639 id=/17 term=0 exit=0

pts/18 2022-06-22 16:04 25550 id=/18 term=0 exit=0

p01vspplmapp01 mpn00221 [~] \$ **who -u** (Same as **who -a** but with less information)

(unknown) :0 2022-06-03 09:51 ? 2006 (:0)

tcuaprd pts/0 2022-06-14 18:15 old 97882 (10.90.20.38)

tcuaprd pts/1 2022-06-03 09:55 old 3344 (10.90.20.36)

tcuaprd pts/2 2022-06-08 15:23 19:44 61500 (10.90.20.38)

mpn00221 pts/3 2022-07-07 09:26 . 61478 (10.90.20.33)

tcuaprd pts/4 2022-06-13 12:31 old 86969 (10.90.20.38)

tcuaprd pts/6 2022-06-15 14:26 old 62662 (10.90.20.36)

tcuaprd pts/8 2022-06-14 19:11 old 102092 (10.90.20.38)

tcuaprd pts/9 2022-06-14 19:13 old 102328 (10.90.20.38)

tcuaprd pts/10 2022-06-14 19:14 old 102516 (10.90.20.38)

tcuaprd pts/11 2022-06-15 11:19 old 47015 (10.90.20.36)

p01vspplmapp01 mpn00221 [~] \$

p01vspplmapp01 mpn00221 [tcuaprd] \$ **pwd** (Present Working Directory Info)

/user/tcuaprd

p01vspplmapp01 mpn00221 [tcuaprd] \$ **cd ~tcuaprd** (Change to another users directory using tilde command)

```
p01vspplmapp01 mpn00221 [tcuaprd] $ pwd
```

```
/user/tcuaprd
```

```
p01vspplmapp01 mpn00221 [tcuaprd] $ cd /home/people/nmm918970/ (change to a directory)
```

```
p01vspplmapp01 mpn00221 [nmm918970] $ pwd
```

```
/home/people/nmm918970
```

```
p01vspplmapp01 mpn00221 [nmm918970] $
```

df command

```
p01vspplmapp01 mpn00221 [tcuaprd] $ df (Display Disk File system Information in Kilo Bytes)
```

```
p01vspplmapp01 mpn00221 [tcuaprd] $ df
```

Filesystem	1K-blocks	Used	Available	Use%	Mounted on
devtmpfs	36962008	0	36962008	0%	/dev
tmpfs	36972432	2816	36969616	1%	/dev/shm
tmpfs	36972432	100560	36871872	1%	/run
tmpfs	36972432	0	36972432	0%	/sys/fs/cgroup
/dev/sda1	25674532	6321600	18287852	26%	/
/dev/sdb1	7455884200	3402546244	4045805384	46%	/data
/dev/sda3	463390448	30895856	431534496	7%	/home
/dev/sdc5	1055838644	258448696	796299828	25%	/data1
/dev/sda4	205366248	482924	203818412	1%	/tmp
tmpfs	7394488	16	7394472	1%	/run/user/484
tmpfs	7394488	0	7394488	0%	/run/user/1001
tmpfs	7394488	0	7394488	0%	/run/user/1002

```
p01vspplmapp01 mpn00221 [tcuaprd] $ df -kh (Display Disk File system Information in Giga Bytes)
```

```
p01vspplmapp01 mpn00221 [tcuaprd] $ df -kh
```

Filesystem	Size	Used	Avail	Use%	Mounted on
devtmpfs	36G	0	36G	0%	/dev
tmpfs	36G	2.8M	36G	1%	/dev/shm
tmpfs	36G	99M	36G	1%	/run
tmpfs	36G	0	36G	0%	/sys/fs/cgroup
/dev/sda1	25G	6.1G	18G	26%	/
/dev/sdb1	7.0T	3.2T	3.8T	46%	/data
/dev/sda3	442G	30G	412G	7%	/home
/dev/sdc5	1007G	247G	760G	25%	/data1
/dev/sda4	196G	472M	195G	1%	/tmp
tmpfs	7.1G	16K	7.1G	1%	/run/user/484
tmpfs	7.1G	0	7.1G	0%	/run/user/1001
tmpfs	7.1G	0	7.1G	0%	/run/user/1002

```
p01vspplmapp01 mpn00221 [tcuaprd] $
```

ls command information.

```
p01vspplmapp01 mpn00221 [~] $ ls (Display the files/directories under the directory)
```

```
clearlocks69611.syslog Jagdish mpn1 oradiag_mpn00221 start_tcuaprd.sh training  
expdpPLMPS.sh mpn nohup.out start_tcuaprd1.sh tecs.out
```

```
p01vspplmapp01 mpn00221 [~] $ ls -l (Display the contents in alphabetically sorted)
```

```
p01vspplmapp01 mpn00221 [~] $ ls -l
```

```
total 440  
-rw-r--r-- 1 mpn00221 dba 11474 Nov 12 2021 clearlocks69611.syslog  
-rwxr-xr-x 1 mpn00221 dba 9913 Apr 6 16:14 expdpPLMPS.sh  
drwxrwxrwx 2 mpn00221 dba 4096 Nov 25 2021 Jagdish  
drwxr-xr-x 2 mpn00221 dba 4096 Jul 6 17:03 mpn  
drwxr-xr-x 2 mpn00221 dba 4096 Jul 6 15:13 mpn1  
-rw-r--r-- 1 mpn00221 dba 258212 Nov 3 2021 nohup.out  
drwxr-xr-x 3 mpn00221 dba 4096 Dec 17 2021 oradiag_mpn00221  
-rw-r--r-- 1 mpn00221 dba 2041 Jul 13 2021 start_tcuaprd1.sh  
-rw-r--r-- 1 mpn00221 dba 2041 Jul 13 2021 start_tcuaprd.sh  
-rw-r--r-- 1 mpn00221 dba 131728 Nov 1 2021 tecs.out  
drwxr-xr-x 2 mpn00221 dba 4096 Jul 6 14:19 training  
p01vspplmapp01 mpn00221 [~] $ ls -lt
```

p01vspplmapp01 mpn00221 [~] \$ **ls -ltr** (Creation date in reverse level, latest file in below level)

```
p01vspplmapp01 mpn00221 [~] $ ls -ltr
total 440
-rw-r--r-- 1 mpn00221 dba 2041 Jul 13 2021 start_tcuaprd.sh
-rw-r--r-- 1 mpn00221 dba 2041 Jul 13 2021 start_tcuaprd1.sh
-rw-r--r-- 1 mpn00221 dba 131728 Nov 1 2021 tecs.out
-rw-r--r-- 1 mpn00221 dba 258212 Nov 3 2021 nohup.out
-rw-r--r-- 1 mpn00221 dba 11474 Nov 12 2021 clearlocks69611.syslog
drwxrwxrwx 2 mpn00221 dba 4096 Nov 25 2021 Jagdish
drwxr-xr-x 3 mpn00221 dba 4096 Dec 17 2021 oradiag_mpn00221
-rwxr-xr-x 1 mpn00221 dba 9913 Apr 6 16:14 expdpPLMPS.sh
drwxr-xr-x 2 mpn00221 dba 4096 Jul 6 14:19 training
drwxr-xr-x 2 mpn00221 dba 4096 Jul 6 15:13 mpn1
drwxr-xr-x 2 mpn00221 dba 4096 Jul 6 17:03 mpn
p01vspplmapp01 mpn00221 [~] $
```

cp command

p01vspplmapp01 mpn00221 [test2] \$ **cp /user/mpn00221/mpn/*** . (When cp it copy the files with latest date & time)

```
p01vspplmapp01 mpn00221 [test2] $ ls -lt
total 1440
-rw-r--r-- 1 mpn00221 dba 1060255 Jul 7 10:56 ppom_user.lst
-rw----- 1 mpn00221 dba 3272 Jul 7 10:56 sample.key
-rwxr-xr-x 1 mpn00221 dba 576 Jul 7 10:56 tasl_activeplmusr.sh
-rwxr-xr-x 1 mpn00221 dba 451 Jul 7 10:56 tasl_fullplmusr.sh
-rwxr-xr-x 1 mpn00221 dba 323 Jul 7 10:56 usersel.sh
-rwxr-xr-x 1 mpn00221 dba 326 Jul 7 10:56 user.sh
-rw-r--r-- 1 mpn00221 dba 109 Jul 7 10:56 accessrun.sh
-rw-r--r-- 1 mpn00221 dba 7806 Jul 7 10:56 info.txt
-rw-r--r-- 1 mpn00221 dba 4100 Jul 7 10:56 input.txt
-rw-r--r-- 1 mpn00221 dba 929 Jul 7 10:56 mailserverinfo.txt
-rw-r--r-- 1 mpn00221 dba 365145 Jul 7 10:56 pperson.lst
p01vspplmapp01 mpn00221 [test2] $
```

p01vspplmapp01 mpn00221 [test2] \$ **cp -p /user/mpn00221/mpn/*** . (Here used option **-p** so files retrieve same date and time of source directory at destination)

```
p01vspplmapp01 mpn00221 [test2] $ cp -p /user/mpn00221/mpn/* .
p01vspplmapp01 mpn00221 [test2] $ ls -lt
total 1440
-rw-r--r-- 1 mpn00221 dba 4100 Apr 28 17:06 input.txt
-rw-r--r-- 1 mpn00221 dba 109 Apr 28 17:06 accessrun.sh
-rwxr-xr-x 1 mpn00221 dba 576 Mar 16 12:08 tasl_activeplmusr.sh
-rw-r--r-- 1 mpn00221 dba 365145 Mar 16 12:08 pperson.lst
-rw-r--r-- 1 mpn00221 dba 1060255 Mar 16 12:07 ppom_user.lst
-rwxr-xr-x 1 mpn00221 dba 451 Mar 16 11:54 tasl_fullplmusr.sh
-rw----- 1 mpn00221 dba 3272 Mar 8 10:57 sample.key
-rwxr-xr-x 1 mpn00221 dba 326 Jan 25 11:25 user.sh
-rw-r--r-- 1 mpn00221 dba 929 Dec 30 2021 mailserverinfo.txt
-rwxr-xr-x 1 mpn00221 dba 323 Dec 20 2021 usersel.sh
-rw-r--r-- 1 mpn00221 dba 7806 Dec 8 2021 info.txt
p01vspplmapp01 mpn00221 [test2] $ pwd
/tmp/testfile/test1/test2
p01vspplmapp01 mpn00221 [test2] $
```

mkdir Command

p01vspplmapp01 mpn00221 [~] \$ **cd /tmp/**

p01vspplmapp01 mpn00221 [tmp] \$ **pwd**

/tmp

p01vspplmapp01 mpn00221 [tmp] \$ **mkdir testfile** (Create a directory)

p01vspplmapp01 mpn00221 [tmp] \$ **ls -ld testfile**

drwxr-xr-x 2 mpn00221 dba 4096 Jul 7 10:50 testfile

p01vspplmapp01 mpn00221 [tmp] \$ **cd testfile/**

p01vspplmapp01 mpn00221 [testfile] \$ **mkdir test1/test2** (Creating directory under a directory)

```

mkdir: cannot create directory 'test1/test2': No such file or directory (Given error for syntax)
p01vspplmapp01 mpn00221 [testfile] $ mkdir -p test1/test2 (Using -p we can create directory under a directory)
p01vspplmapp01 mpn00221 [testfile] $ pwd
/tmp/testfile
p01vspplmapp01 mpn00221 [testfile] $ cd test1/test2
p01vspplmapp01 mpn00221 [test2] $ pwd
/tmp/testfile/test1/test2
p01vspplmapp01 mpn00221 [testfile] $ touch 1.txt (Creating a file)
p01vspplmapp01 mpn00221 [testfile] $ ls -l
total 0
-rw-r--r-- 1 mpn00221 dba 0 Jul 7 10:50 1.txt
p01vspplmapp01 mpn00221 [test2] $ pwd
/tmp/testfile/test1/test2
p01vspplmapp01 mpn00221 [test2] $

```

mv Command

Basically, mv renames source file to target file. If there is a file with the same name as target file, it will be overwritten. mv works for directories in a similar fashion.

We can rename the directory with mv command.

rm command

rm file(s) – delete file(s).

rmdir directories – delete one or more empty directories. Also can user rm -f

rm -r directories – can be used to delete non empty directories.

rm -i <file name> it will ask before remove the file, as below.

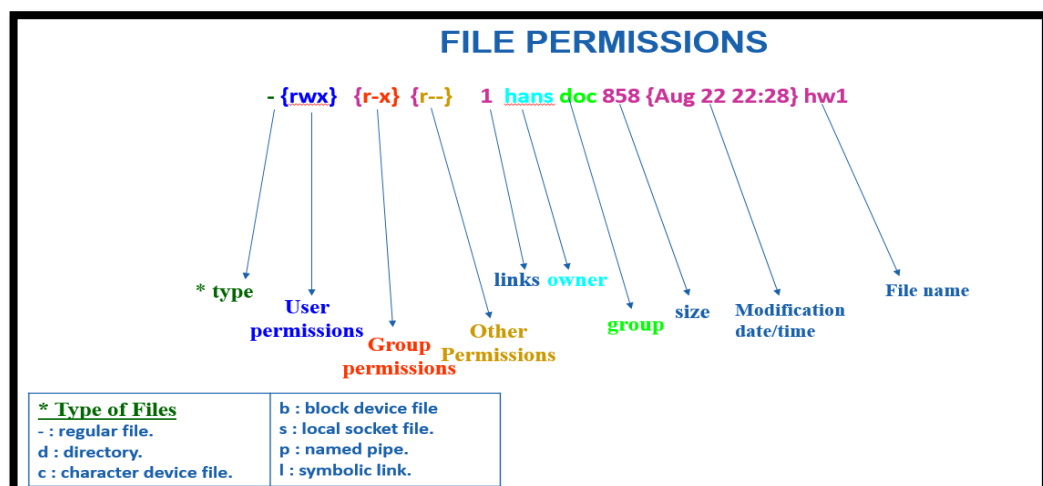
```
p01vspplmapp01 mpn00221 [test2] $ rm -i usersel.sh
```

rm: remove regular file 'usersel.sh'?

!!!WARNING!!! This will DELETE EVERYTHING in that directory!!!

You cannot recover your files after you removed them (unlike Windows OS).

File Permissions

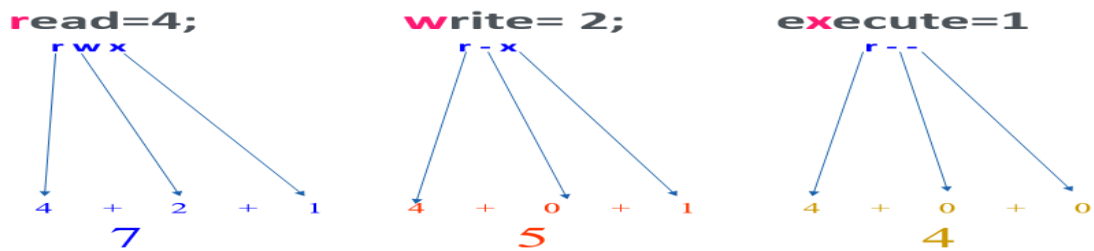


NUMERICAL PERMISSIONS

#	Sum	<u>rwX</u>	Permission
7	4(r) + 2(w) + 1(x)	r w x	read, write and execute
6	4(r) + 2(w)	r w -	read and write
5	4(r) + 1(x)	r - x	read and execute
4	4(r)	r - -	read only
3	2(w) + 1(x)	- w x	write and execute
2	2(w)	- w -	write only
1	1(x)	- - x	execute only
0	0	- - -	none

- There are three types of file access supported by UNIX.
 - r** – read, view the contents of a file or a directory
 - w** – write, edit file/directory contents
 - x** – execute, run executable file

PERMISSIONS



chmod command

```
p01vspplmapp01 mpn00221 [test2] $ ls -lt user.sh
-rwxr-xr-x 1 mpn00221 dba 326 Jan 25 11:25 user.sh
p01vspplmapp01 mpn00221 [test2] $ chmod 777 user.sh
p01vspplmapp01 mpn00221 [test2] $ ls -lt user.sh
-rwxrwxrwx 1 mpn00221 dba 326 Jan 25 11:25 user.sh
p01vspplmapp01 mpn00221 [test2] $ chmod 700 user.sh
p01vspplmapp01 mpn00221 [test2] $ ls -lt user.sh
-rwx----- 1 mpn00221 dba 326 Jan 25 11:25 user.sh
p01vspplmapp01 mpn00221 [test2] $ chmod 644 user.sh
p01vspplmapp01 mpn00221 [test2] $ ls -lt user.sh
-rw-r--r-- 1 mpn00221 dba 326 Jan 25 11:25 user.sh
p01vspplmapp01 mpn00221 [test2] $
```

Syntax

- chmod [-R] mode file ...
- chmod [-R] [ugoa]{+|-|=[rwxXstl]} file ...

Description

- Changes the permissions mode of a file or directory.

Examples

- chmod 444 file1
- chmod ugo+rw p*.c
- chmod 700 mydir
- chmod u+/-rwx <file name>
- chmod g+/-rwx <file name>
- chmod o+/-rwx <file name>

grep command

```
p01vspplmapp01 mpn00221 [~] $ cat /etc/passwd | grep mpn00221
mpn00221:x:1002:1000:PDM Nandakumar M P:/user/mpn00221:/bin/bash
p01vspplmapp01 mpn00221 [~] $ grep mpn00221 /etc/passwd
mpn00221:x:1002:1000:PDM Nandakumar M P:/user/mpn00221:/bin/bash
p01vspplmapp01 mpn00221 [~] $ egrep "mpn|nmm" /etc/passwd
mpn00221:x:1002:1000:PDM Nandakumar M P:/user/mpn00221:/bin/bash
nmm918970:x:1007:1000:PDM Nikhil Marathe:/user/nmm918970:/bin/bash
p01vspplmapp01 mpn00221 [~] $ cat /etc/passwd | egrep "mpn|nmm"
mpn00221:x:1002:1000:PDM Nandakumar M P:/user/mpn00221:/bin/bash
nmm918970:x:1007:1000:PDM Nikhil Marathe:/user/nmm918970:/bin/bash
p01vspplmapp01 mpn00221 [~] $ egrep "MPN|NMM" /etc/passwd
p01vspplmapp01 mpn00221 [~] $ #No Output because not matching, so use -i option
p01vspplmapp01 mpn00221 [~] $ egrep -i "MPN|NMM" /etc/passwd
mpn00221:x:1002:1000:PDM Nandakumar M P:/user/mpn00221:/bin/bash
nmm918970:x:1007:1000:PDM Nikhil Marathe:/user/nmm918970:/bin/bash
```

find command

find - find files

SYNOPSIS

find path-name-list [expression]

find . -name <string> -print

find . \(-name <string1> -o -name <string2> \) -print

find with execute command

find . -name <string> -print -exec <command> {} \;

find . -name <string> -print -exec ls -lt {} \;

p01vspplmapp01 mpn00221 [testfile] \$ cd /tmp/testfile/

p01vspplmapp01 mpn00221 [testfile] \$ pwd

/tmp/testfile

p01vspplmapp01 mpn00221 [testfile] \$

p01vspplmapp01 mpn00221 [testfile] \$ **find . -name user.sh**

./test3/test2/user.sh

p01vspplmapp01 mpn00221 [testfile] \$ **find . -name user.sh -exec ls -lt {} \;**

-rw-r--r-- 1 mpn00221 dba 326 Jan 25 11:25 ./test3/test2/user.sh

p01vspplmapp01 mpn00221 [testfile] \$ **find . -name user.sh -exec rm -f {} \;** -print

./test3/test2/user.sh

p01vspplmapp01 mpn00221 [testfile] \$ **find . -name user.sh -exec ls -lt {} \;** (File removed so not found)

p01vspplmapp01 mpn00221 [testfile] \$

To find the size of the file

p01vspplmapp01 mpn00221 [testfile] \$ find . -size 6M -exec ls -lt {} \;

p01vspplmapp01 mpn00221 [testfile] \$ find . -size +10M -size -15M -exec ls -lt {} \;

<https://www.tecmint.com/35-practical-examples-of-linux-find-command/>

The above link gives more examples of find command.

sed command (Stream Editor)

(vi the following line)

Mary had a little lamb

Mary fried a lot of spam

Jack ate a Spam sandwich

Jill had a ham spamwich

sed 's/lamb/ham/g' poem.txt

sed '2,3d' poem.txt

Search and replace:

sed 's/lamb\$/ham/g' poem.txt > new.file

%s/<search string>/<replace string>/g

Substitution sed 's/<oldstring>/<newstring>/g' <file>

Deletion sed '<start>,<end>d' <file>

cat passwd | sed 's/<old string>/<New string>/'

cat passwd | sed 's/<old string>/<New string>/' >filename -

(This will copy the file into new file name)

cut Command

```
p01vspplmapp01 mpn00221 [mpn] $ cat input.txt
nsd653117~ERC_Designers_Group~D401~00~D~
sde532422~ERC_Manager_Group~D401~01~D~
ssd775676~ERC_Review_Group~D401~02~D~
mpn765457~ERC_Release_Group~D401~03~D~
p01vspplmapp01 mpn00221 [mpn] $ cat input.txt | cut -c 1-9
nsd653117
sde532422
ssd775676
mpn765457
p01vspplmapp01 mpn00221 [mpn] $ cat input.txt | cut -d~ -f2
ERC_Designers_Group
ERC_Manager_Group
ERC_Review_Group
ERC_Release_Group
p01vspplmapp01 mpn00221 [mpn] $
```

awk command

cat /etc/passwd | awk -F: '{print \$1 " " \$3 " " \$5}'

cat /etc/passwd | awk -F: '{printf "%s %d %s \n", \$1,\$3,\$5}'

cat /etc/passwd | awk -F: '{printf "%10s %10d %30s \n", \$1,\$3,\$5}'

cat /etc/passwd | awk -F: '{printf "%-10s %-6d %-30s \n", \$1,\$3,\$5}'

ls -lt | awk '{print \$9}'

ps -ef | grep omfprod | awk '{print \$1,\$2}'

Server OS Details.

p01vspplmapp01 mpn00221 [mpn] \$ **uname -a**

Linux p01vspplmapp01 4.12.14-122.37-default #1 SMP Sun Sep 6 05:00:36 UTC 2020 (fe8cacf) x86_64 x86_64
x86_64 GNU/Linux

p01vspplmapp01 mpn00221 [mpn] \$ **cat /etc/SuSE-release**

SUSE Linux Enterprise Server 12 (x86_64)

VERSION = 12

PATCHLEVEL = 5

This file is deprecated and will be removed in a future service pack or release.

Please check /etc/os-release for details about this release.

p01vspplmapp01 mpn00221 [mpn] \$

Option Description

uname

-a Displays all information.

-r Displays the kernel version.

-m Displays the hardware (CPU) type.

-n Displays the system's host name.

-s Displays the operating system name (equivalent to uname).

-p Displays the system's processor type.

-v Prints the Linux version, that is, when it was compiled

Some file names are special:

- / The root directory (not to be confused with the root user)
- . The current directory
- .. The parent (previous) directory

- ~ My home directory

Examples:

- ./a same as a
- ../jane/x go up one level then look in directory jane for x

Other Commands

ps -ef find out the current processes
 kill kill one process
 zip zip the files
 unzip unzip the zipped files
 tar -cvf collect group of files or directories
 tar -xvf extract the tarred files
 tar -tvf Display the tarred files.

Monitoring and diagnosis of system performance (Admin Related)

top
 vmstat
 iostat
 netstat -nap / netstat -ntlp / netstat -nulp

Example of Small Shell Script

Details of the shell

1. The shell script begins with the line "#!/bin/sh" .
2. Usually "#" denotes the start of a comment, but #! is a special combination that tells UNIX to use the Bourne shell (sh) to interpret this script.
3. The #! must be the first two characters of the script.
4. The arguments passed to the script can be accessed through \$1, \$2, \$3 etc.
5. \$* stands for all the arguments, and \$# for the number of arguments.
6. The process number of the shell executing the script is given by \$\$. the read number statement assigns keyboard input to the variable number.

Shells

```
# This script displays the date, time, username and
current directory.
vi s1.sh ( check the permission)
echo "Date and time is:" `date`
echo
echo -e "Your username is: `whoami` \n"
echo -e "Your current directory is: `c`"
pwd
-----
Date you can manipulate with date '+%d%m%y'
```

```
#!/bin/ksh
# this is a comment
echo "The number of arguments is $#"
```

```
echo "The arguments are $*"
echo "The first is $1"
```

```
echo "My process number is $$"
echo "Enter a number from the keyboard: "
read number
echo "The number you entered was $number"
See the permission of the shell, is it able to execute.
(See the Output)
```

To Create a Directory.

```
if [ ! -d /tmp/test_dir ]
then
    mkdir -p /tmp/test_dir
    chmod 777 /tmp/test_dir
    ls -ld /tmp/test_dir
else
    echo "File exists"
fi
```